Magisterij notes :

SC : $H = \sum_k (c_k \;\; c_{-k}^+) \underbrace{\begin{pmatrix} -\xi_k & -\Delta_k^* \\ -\Delta_k & \xi_k \end{pmatrix}}_{\mathcal{H}_k} \begin{pmatrix} c_k^+ \\ c_{-k} \end{pmatrix} \underbrace{\left(-\sum_{kk'} V_{kk'} \langle c_k^+ c_{-k}^+ \rangle \langle c_{-k'} c_{k'} \rangle + \sum_k \xi_k \right)}_{k_0 \text{ - energy shift}}$

$\Delta_k = \sum_{k'} V_{kk'} \langle c_{-k'} c_{k'} \rangle \quad , \quad \Delta_k^* = \sum_{k'} V_{kk'}^* \langle c_{k'}^+ c_{-k'}^+ \rangle$

diagonalizacija :

① Bogoljubov :

$\alpha_k = u_k c_k - v_k c_{-k}^+ \qquad \alpha_{-k} = u_k c_{-k} + v_k c_k^+$

$\alpha_k^+ = u_k c_k^+ - v_k c_{-k} \qquad \alpha_{-k}^+ = u_k c_{-k}^+ + v_k c_k$ | Bogoljubov

inverse

$c_k = u_k \alpha_k + v_k \alpha_{-k}^+ \qquad c_{-k} = u_k \alpha_{-k} - v_k \alpha_k^+$

$c_k^+ = u_k \alpha_k^+ + v_k \alpha_{-k} \qquad c_{-k}^+ = u_k \alpha_{-k}^+ - v_k \alpha_k$

$\cdot \; u_k, v_k \in \mathbb{R}, \quad u_k = u_{-k}, \quad v_k = v_{-k} \quad \rightarrow$ CCR : $\boxed{u_k^2 + v_k^2 = 1}$

je to $K = H - \mu N$!

$H = \sum_k \xi_k (c_k^+ c_k + c_{-k}^+ c_{-k}) - \bar{V} \sum_{kk'}' \underbrace{c_{-k'}^+ c_{k'}^+ c_k c_{-k}}_{\substack{\| \\ b_k^+ b_{k'}}}$ / MF : def: $b_k \equiv c_{-k} c_k$, $b_k^+ = c_k^+ c_{-k}^+$

$b_k^+ b_{k'} \rightarrow b_k = \langle b_k \rangle + \delta b_k = F_k + \delta b_k$

$b_k^+ = \langle b_k^+ \rangle + \delta b_k^+ = F_k^+ + \delta b_k^+$

$\approx F_k^* F_{k'} + F_k^+ \delta b_{k'} + F_{k'} \delta b_k^+ = \cancel{F_k^* F_{k'}} - \cancel{F_k^* F_{k'}} + F_k^+ b_{k'} + F_{k'} b_k^+ - F_k^* F_{k'}$

$H^{MF} = \sum_k \xi_k (c_k^+ c_k + c_{-k}^+ c_{-k}) - \sum_k \left[\left(\sum_{k'} \bar{V} \langle c_{-k'}^+ c_{k'}^+ \rangle \right) c_k c_{-k} + \left(\sum_{k'} \bar{V} \langle c_{k'} c_{-k'} \rangle \right) c_{-k}^+ c_k^+ \right] + \bar{V} \underbrace{\sum_{kk'} \langle \; \rangle \langle \; \rangle}_{\| \; \frac{\Delta^2}{\bar{V}}} =$

$cc^+ + c^+c = 1 \qquad \| \; -c_k c_k^+ + 1$

def: $\Delta_k = \sum_{k'} V_{kk'} \langle c_{k'} c_{-k'} \rangle \simeq \Delta$ za $V_{kk'} \simeq \bar{V}$

$= \sum_k (c_k \;\; c_{-k}^+) \begin{pmatrix} -\xi_k & -\Delta_k^* \\ -\Delta_k & \xi_k \end{pmatrix} \begin{pmatrix} c_k^+ \\ c_{-k} \end{pmatrix} + \left(\frac{\Delta^2}{\bar{V}} + \sum_k \xi_k \right) =$ (drugačić dmeš $c_k c_k^+$)

$= \sum_k (c_k^+ \;\; c_{-k}) \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^* & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^+ \end{pmatrix} + \left(\frac{|\Delta|^2}{\bar{V}} + \sum_k \xi_k \right)$

? minus?

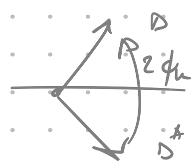
Bogoljubov :

$= \sum_k (\alpha_k^+ \;\; \alpha_{-k}) \underbrace{\begin{pmatrix} u & -v \\ v & u \end{pmatrix} \begin{pmatrix} \xi & -\Delta \\ -\Delta^* & -\xi \end{pmatrix} \begin{pmatrix} u & v \\ -v & u \end{pmatrix}} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^+ \end{pmatrix}$

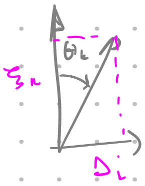
$\begin{pmatrix} u & -v \\ v & u \end{pmatrix} \begin{pmatrix} \xi u - \Delta v & \xi v + \Delta u \\ \Delta^* u + \xi v & \Delta^* v - \xi u \end{pmatrix} = \begin{pmatrix} \xi u^2 + uv(\Delta^* - \Delta) - \xi v^2 & 2\xi uv + \Delta u^2 - \Delta^* v^2 \\ - \| - & \xi v^2 - \xi u^2 + uv(\Delta + \Delta^*) \end{pmatrix}$

$u^2 + v^2 = 1 \rightarrow \quad u = \cos\frac{\theta_k}{2} \qquad v = \sin\frac{\theta_k}{2} e^{i\phi_k}$

$\Rightarrow \quad \xi_k 2 \sin\left(\frac{\theta_k}{2}\right) \cos\left(\frac{\theta_k}{2}\right) = +|\Delta| \left(\cos\left(\frac{\theta_k}{2}\right)^2 - \sin\left(\frac{\theta_k}{2}\right)^2 \right) \qquad \Delta^* e^{i2\phi_k} = \Delta$



$\sin(\theta_k)\,\xi_k = |\Delta|\cos\theta_k \Rightarrow$ $\boxed{\tan\theta_k = \frac{|\Delta_k|}{\xi_k}}$ in $\boxed{\begin{array}{l}\mu_k = \cos\frac{\theta_k}{2}\\ \nu_k = \sin\frac{\theta_k}{2}\end{array}}$



$\to \cos\theta_k = \frac{\xi}{\sqrt{\Delta^2+\xi^2}}$, $\sin\theta_k = \frac{\Delta}{\sqrt{\xi^2+\Delta^2}} \leftarrow (*)$

$\mu_k = \cos\frac{\theta_k}{2} = \sqrt{\frac{1+\cos\theta_k}{2}}$ $\to$ $\boxed{\mu_k^2 = \frac{E_k+\xi_k}{2E_k}, \quad \nu_k^2 = \frac{E_k-\xi_k}{2E_k}}$

$$H = \sum_k E_k\alpha_k^\dagger\alpha_k + (\xi_k - E_k) - \frac{\Delta^2}{V} \leftarrow$$

$+$ GS: $\alpha_k|BCS\rangle = 0 \Rightarrow$ $\boxed{|BCS\rangle = \prod_k(\mu_k + \nu_k c_k^\dagger c_{-k}^\dagger)|0\rangle}$

(2) Pauli | Anderson:

p-tolma: $\Delta_k = \frac{V}{2}\sum\sin\theta_k = V\sum\langle c_k c_{-k}\rangle = V\sum\frac{\langle X_k\rangle}{2} = V\sum\langle S_x\rangle$

Self consistency: $\langle\alpha_k^\dagger\alpha_k\rangle = f_k = \frac{1}{e^{\beta E_k}+1} = \frac12 - \frac12\,\mathrm{th}\left(\frac12\beta E_k\right)$

$\Rightarrow \langle c_k^\dagger c_k\rangle = \frac12 - \frac{\xi_k}{2E_k}\mathrm{th}\left(\frac12\beta E_k\right)$

$\langle c_{-k\sigma}c_{k\sigma}\rangle = -\frac{\sigma\Delta_k}{2E_k}\mathrm{th}\left(\frac12\beta E_k\right)$

$|\xi_k| < \omega_D$

$T=0$ $\quad (*)$

by def: $\Delta_k = \sum_{k'} V_{kk'}\langle c_{k'}c_{k'}\rangle \simeq \frac{V}{2}\sum_{k'}\sin\theta_{k'} = \frac{V}{2}\sum_{k'}{}'\frac{\Delta_{k'}}{\sqrt{\Delta_k^2+\xi_k^2}}$

per spin

$\to$ $\boxed{\frac{1}{V} = \frac12\int_{-\omega_D}^{\omega_D}\frac{\frac12 g(\xi)}{\sqrt{\Delta^2+\xi^2}}d\xi}$ $\to$ $\boxed{\Delta \simeq 2\omega_D e^{-\frac{2}{g(\epsilon_F)V}}}$

by def: $\langle c_{-k}c_k\rangle = \langle(\mu_k\alpha_{-k} - \nu_k\alpha_k^\dagger)(\mu_k\alpha_k + \nu_k\alpha_{-k}^\dagger)\rangle =$

$= \mu_k\nu_k\left(1 - \langle\alpha_k^\dagger\alpha_k\rangle - \langle\alpha_{-k}^\dagger\alpha_{-k}\rangle\right) = \frac12\sin\theta_k\cdot\mathrm{th}\left(\frac{\beta E_k}{2}\right)$

$\cos\frac{\theta_k}{2}\sin\frac{\theta_k}{2} = \frac12\sin\theta_k$

$1 - f_k - f_{-k} = 1 - \frac{2}{e^{\beta E_k}+1}$

• od kje self. cons.: (1) Δ arbitraren, ne vemo kaj je $\langle cc\rangle$, odvisno od vpliva C.P.

(2) Ampak Δ je vključena v $H^{MF}(\Delta)$!

loop: $\Delta \to H^{MF}(\Delta) \to |\text{stanje}(\Delta)\rangle \to \langle c_k c_k\rangle \to \Delta'$ repeat.

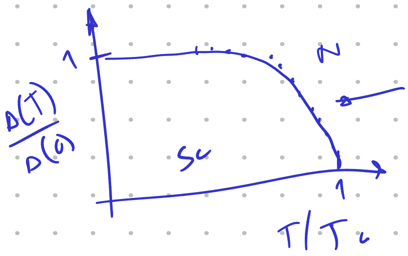
→ vsak C.P. čuti kolektivni vpliv ostalih parov in si ustvari svoj prispevek k $\langle cc\rangle$, kar prispeva k ostalim, ki spet spremenijo stanje, kar spremeni kako stanje C.P. .

$T>0$: vemo $\langle c_k c_{-k}\rangle = \frac{1}{2}\sin\theta_k \,\mathrm{th}\left(\frac{\beta E_k}{2}\right) = \frac{\Delta_k}{2E_k}\,\mathrm{th}\left(\frac{\beta E_k}{2}\right)$

$$\Rightarrow \Delta \simeq \frac{V}{2}\sum_k \sin\theta_k\, \mathrm{th}\left(\frac{\beta E_k}{2}\right) = \frac{V\Delta}{2}\sum_k{}' \frac{\mathrm{th}\left(\frac{\beta}{2}\sqrt{\Delta^2+\xi_k^2}\right)}{\sqrt{\Delta^2+\xi_k^2}} \Rightarrow$$

$$\Rightarrow \boxed{\frac{1}{V} \simeq \frac{1}{2}\int_{-\omega_D}^{\omega_D} d\xi\, \frac{\frac{1}{2} g(\epsilon_F)\, \mathrm{th}\left(\frac{\beta}{2}\sqrt{\Delta^2+\xi^2}\right)}{\sqrt{\Delta^2+\xi^2}}} \rightarrow \Delta(T) = \ldots$$

per spin!



$\Delta(T)=0$ numer.

numerično: (1) $$\Delta'_{n+1} = \frac{g(\epsilon_F)V}{2}\cdot \Delta_n \cdot \sum_{\xi=0}^{\omega_D} \overset{d\xi}{\left(\frac{\omega_D}{N}\right)} \frac{\mathrm{th}\left(\frac{\beta}{2}\sqrt{\Delta_n^2+\xi^2}\right)}{\sqrt{\Delta_n^2+\xi^2}}$$

(2) $\Delta_{n+1} = (1-a)\Delta_n + a\Delta'_{n+1}$ (stabilnost)!

(3) če $|\Delta_{n+1}-\Delta_n| < \varepsilon$ stop..

efektivno imamo iteracijo: $\Delta_{n+1} = A(T)\Delta_n$; če $A(T)>1 \Rightarrow \Delta>0$

če $A(T)<1 \Rightarrow \Delta=0$

pri $T=T_c \rightarrow A(T)=1$

konvergenca počasna okoli $A=1$!

2) EXCITON INSULATOR:

i, j mesti, a, b orbitali,

→ Coulomb, interband

$$\to H = H_0 + H_V = \sum_{\substack{ij\\ \gamma=a,b}} \left(\varepsilon^{(\gamma)}\delta_{ij} + t_{ij}^{(\gamma)}\right) a_i^\dagger a_j + V\sum_j a_j^\dagger a_j b_j^\dagger b_j =$$

$$= \sum_k \varepsilon_a(k)\, a_k^\dagger a_k + \varepsilon_b(k)\, b_k^\dagger b_k + \underbrace{V\sum_j a_j^\dagger a_j b_j^\dagger b_j}$$

$\varepsilon_a(k)$, $\varepsilon_b(k)$ } non-interacting

lokalna interakcija, on site, lahko spremenimo!

MF za H_V:

$$a_j^\dagger a_j b_j^\dagger b_j \simeq \underbrace{\langle a_j^\dagger a_j\rangle}_{n_a} b_j^\dagger b_j + \underbrace{\langle b_j^\dagger b_j\rangle}_{n_b} a_j^\dagger a_j - n_a n_b \quad \leftarrow \text{Hartree}$$

$$- \underbrace{\langle a_j^\dagger b_j\rangle}_{\phi} b_j^\dagger a_j - \underbrace{\langle (a_j^\dagger b_j)^\dagger\rangle}_{\phi^*} a_j^\dagger b_j + \langle a_j^\dagger b_j\rangle\langle b_j^\dagger a_j\rangle \leftarrow \text{Fock/exchange}$$

$a^\dagger a b^\dagger b = -a^\dagger b^\dagger a b = (-1)^3 a^\dagger b\, b^\dagger a$

od fermionov

elektroni (density $\hat{n}_b$) čutijo average density v drugem bandu.

$$H_V^{MF} = V n_b \sum_j a_j^\dagger a_j + V n_a \sum_j b_j^\dagger b_j - V\phi\sum_j b_j^\dagger a_j - V\phi^*\sum_j a_j^\dagger b_j + NV\left(|\phi|^2 - n_a n_b\right)$$

→ FT: $a_j^\dagger = \frac{1}{\sqrt{N}}\sum_k e^{ikR_j} a_k^\dagger$ etc.

$$\to n_a = \langle a_i^\dagger a_i\rangle = \frac{1}{N}\sum_{kk'} e^{i(k-k')R_i}\langle a_k^\dagger a_{k'}\rangle = \frac{1}{N}\sum_k \langle a_k^\dagger a_k\rangle$$

$$\to \phi = \langle a_j^\dagger b_j\rangle = \frac{1}{N}\sum_k \langle a_k^\dagger b_k\rangle \to \Delta \hat{=} -\frac{V}{N}\sum_k \langle a_k^\dagger b_k\rangle, \quad \Delta^* = -\frac{V}{N}\sum_k\langle b_k^\dagger a_k\rangle$$

$$\Delta = -\frac{1}{N}\sum_k \underbrace{V(k-k')}\langle a_k^\dagger b_k\rangle.$$

$$V(\vec{q}) = \int e^{i\vec{q}\vec{r}} V(\vec{r})\, d\vec{r} \qquad N\frac{NV}{N} = N^2\bar{V}$$

na koncu:

$$H^{MF} = \sum_k \left(\left[\varepsilon_a(k) + V n_b\right] a_k^\dagger a_k + \left[\varepsilon_b(k) + V n_a\right] b_k^\dagger b_k + \Delta^* a_k^\dagger b_k + \Delta b_k^\dagger a_k\right) + \left(\frac{|\Delta|^2}{\bar{V}} - \bar{V} N^2 n_a n_b\right)$$

$$\underbrace{\left(\bar{V} = \frac{V}{N}\right)}$$

enako kot pri BCS.

$$H^{MF} = \sum_k \begin{pmatrix} a_k^\dagger & b_k^\dagger \end{pmatrix} \begin{pmatrix} \overbrace{\varepsilon_a + V n_b}^{\bar{\varepsilon}_a} & \Delta^* \\ \Delta & \underbrace{\varepsilon_b + V n_a}_{\bar{\varepsilon}_b} \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix} + \text{const.}$$

bands: $$E_\pm = \frac{\bar{\varepsilon}_a + \bar{\varepsilon}_b}{2} \pm \sqrt{\left(\frac{\bar{\varepsilon}_a - \bar{\varepsilon}_b}{2}\right)^2 + \Delta^2}$$

self consistency ET:

$$\begin{pmatrix}a_k\\ b_k\end{pmatrix}=\begin{pmatrix}u&v\\ -v&u\end{pmatrix}\begin{pmatrix}\alpha_k\\ \beta_k\end{pmatrix}$$ } samo mix med dvema bandoma

$$\begin{pmatrix}a_k^\dagger & b_k^\dagger\end{pmatrix}=\begin{pmatrix}\alpha_k^\dagger & \beta_k^\dagger\end{pmatrix}\begin{pmatrix}u&-v\\ v&u\end{pmatrix}$$

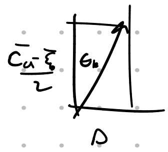
$$\to \mathcal{H}_k=\begin{pmatrix}u&-v\\ v&u\end{pmatrix}\begin{pmatrix}\bar\varepsilon_a&\Delta^*\\ \Delta&\bar\varepsilon_b\end{pmatrix}\begin{pmatrix}u&v\\ -v&u\end{pmatrix}=$$

$$=\begin{pmatrix}u&-v\\ v&u\end{pmatrix}\begin{pmatrix}\bar\varepsilon_a u-\Delta^* v & \bar\varepsilon_a v+\Delta^* u\\ \Delta u-v\bar\varepsilon_b & \Delta v+\bar\varepsilon_b u\end{pmatrix}=\begin{pmatrix}\bar\varepsilon_a u^2-\Delta^* uv+v^2\bar\varepsilon_b-\Delta uv & (\bar\varepsilon_a-\bar\varepsilon_b)uv+\Delta^* u^2-\Delta v^2\\ -\|-\ ^* & \bar\varepsilon_a v^2+\bar\varepsilon_b u^2+\Delta^* uv+\Delta uv\end{pmatrix}$$

$$1=\{a_k,a_k^\dagger\}=\{u\alpha+v\beta,\ u^*\alpha^\dagger+v^*\beta^\dagger\}=u^2+v^2\to u_k^2+v_k^2=1\Rightarrow u_k=\cos\frac{\theta_k}{2}$$
$$\to v_k=\sin\frac{\theta_k}{2}e^{i\varphi_k}$$

$$\Rightarrow \frac{(\bar\varepsilon_a-\bar\varepsilon_b)}{2}\sin\theta_k+\Delta\cos\theta_k=0\to \tan\theta_k=\frac{-2\Delta}{\bar\varepsilon_a-\bar\varepsilon_b}$$

$\hookrightarrow u_k v_k=\frac{\sin\theta_k}{2}$



$\Delta^*=\Delta e^{i\varphi_k}$

$$\Rightarrow \langle a_k^\dagger b_k\rangle=\langle(u_k\alpha_k^\dagger+v_k\beta_k^\dagger)(-v_k\alpha_k+u_k\beta_k)\rangle=u_kv_k\left(\langle\beta_k^\dagger\beta_k\rangle-\langle\alpha_k^\dagger\alpha_k\rangle\right)=$$
$$=-\frac{1}{2}\sin\theta_k\left(-f_\alpha+f_\beta\right)$$

$$\Delta=+V\sum_k\langle a_k^\dagger b_k\rangle=-\frac{V}{2}\sum_k\frac{\Delta}{\sqrt{\underbrace{\left(\frac{\bar\varepsilon_a-\bar\varepsilon_b}{2}\right)^2}_{\xi_k^2}+\Delta^2}}\cdot\left(\underset{f(E_+)}{f_\alpha}-f_\beta\right)$$

$$\to \frac{1}{V}=-\frac{1}{2}\sum_k\frac{1}{\sqrt{\xi_k^2+\Delta^2}}\cdot\left(f(E_+)-f(E_-)\right)$$ ✓

ker je $H\begin{pmatrix}u_k\\ -v_k\end{pmatrix}=E_+\begin{pmatrix}u_k\\ -v_k\end{pmatrix}\to H=\sum_k E_+\alpha_k^\dagger\alpha_k+E_-\beta_k^\dagger\beta_k+$ skalar

$\langle\alpha_k^\dagger\alpha_k\rangle=f(E_+)$!

če $\bar\varepsilon_a=-\bar\varepsilon_b$ aka: $n_a=n_b$

potem $E_\pm=0\pm\sqrt{\xi_k^2+\Delta^2}\to$ se nanaša na BCS:

$$f(E_-)-f(E_+)=f(-E)-f(E)=\frac{1}{e^{+\beta(-E-\mu)}+1}-\frac{1}{e^{\beta(E-\mu)}+1}=1-2f(E)=\operatorname{th}\left(\frac{\beta E}{2}\right)$$ ✓

$E_-=-\sqrt{\xi_k^2+\Delta^2}$

$E_+=+\sqrt{\xi_k^2+\Delta^2}$

$\hookrightarrow f(-E)=1-f(E)$

numerika za EI gap:

- $E_\pm = E_\pm(n_a, n_b, \Delta) = \dfrac{\varepsilon_a(k)+\varepsilon_b(k)+V(n_a+n_b)}{2} \pm \left(\left[\dfrac{\varepsilon_a(k)+\varepsilon_b(k)+V(n_a+n_b)}{2}\right]^2 + |\Delta|^2\right)^{1/2}$

- $f(E_\pm, \mu) = \dfrac{1}{\exp(\beta(E_\pm-\mu))+1}$ ← $n = n_a+n_b = \text{const.} \Rightarrow \mu \neq \text{const.}$

  $\hookrightarrow N = \int_{-\infty}^{\infty} d\varepsilon_a f(E_+) g(\varepsilon_a) + \int_{-\infty}^{\infty} d\varepsilon_b f(E_-) g(\varepsilon_b) = \text{const.} \Rightarrow \mu^{(\ell)} = \dots$

- gap: $\Delta^{(\ell+1)} = \dfrac{V}{2}\Delta^{(\ell)} \sum_k \dfrac{1}{\left(\Delta^{(\ell)2} + \left[\dfrac{\varepsilon_a - \varepsilon_b + V(n_a^{(\ell)} - n_b^{(\ell)})}{2}\right]^2\right)} \left[f(E_-^{(\ell)}) - f(E_+^{(\ell)})\right].$

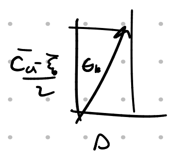
$$\underline{\text{init}}\quad (\Delta, n_a, n_b) \to (\bar\varepsilon_a, \bar\varepsilon_b) \to (E_+, E_-) \xrightarrow[\text{dvranimo } n = \text{const.}]{} \mu \xrightarrow{\text{gap eq.}} (\Delta_{\text{nov}}, n_a^{\text{nov}}, n_b^{\text{nov}})$$

n_a, n_b updated with:

$$n_a = \frac{1}{N}\sum_k \left(u_k^2 f(E_+) + v_k^2 f(E_-)\right) \qquad u_k, v_k = u_k, v_k(\Delta, n_a, n_b)$$
$$n_b = \frac{1}{N}\sum_k \left(v_k^2 f(E_+) + u_k^2 f(E_-)\right)$$

$\tan\theta_k = \dfrac{-2\Delta}{\bar\varepsilon_a - \bar\varepsilon_b}$

$\hookrightarrow u_k v_k = -\dfrac{\sin\theta_k}{2}$



$u_k v_k = -\frac{1}{2}\sin\theta_k$

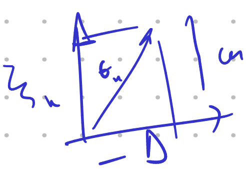
$\tan\theta_k = -\dfrac{\Delta}{\xi_k}$

def $\boxed{\begin{aligned}\xi_k &= \frac{\bar\varepsilon_a - \bar\varepsilon_b}{2}\\ \eta_k &= \frac{\bar\varepsilon_a + \bar\varepsilon_b}{2}\end{aligned}}$ $\boxed{\begin{aligned}\eta_k &= \eta_k(n_a, n_b)\\ \xi_k &= \xi_k(n_a, n_b)\end{aligned}}$

$\rightarrow u_k^2 + v_k^2 = 1 \qquad -\dfrac{\sin\theta_k}{2} = \cos\theta_k \cdot \dfrac{\Delta}{2\xi_k} = u_k v_k$

$$u_k^2 = \cos^2\frac{\theta_k}{2} = \frac{1}{2}(1+\cos\theta_k) = \frac{1}{2}\left(1+\frac{\xi_k}{\sqrt{\Delta^2+\eta_k^2}}\right) = \frac{1}{2}\left(1+\frac{\xi_k}{\sqrt{\Delta^2+\xi_k^2}}\right) = u_k^2$$

$$v_k^2 = \frac{1}{2}\left(1-\frac{\xi_k}{\sqrt{\Delta^2+\xi_k^2}}\right)$$



$\cos\theta_k = \dfrac{\xi_k}{\sqrt{\Delta^2+\xi_k^2}}$

$$n_a = \frac{1}{N}\sum_k \langle a_k^\dagger a_k\rangle = \langle (u_k\alpha_k^\dagger + v_k\beta_k^\dagger)(u_k\alpha + v\beta)\rangle = \frac{1}{N}\sum_k \left[u_k^2 f(E_+) + v_k^2 f(E_-)\right] = n_a$$

$$\langle\alpha^\dagger\alpha\rangle = f(E_+)$$

$$\frac{1}{N}\sum_k \left[v_k^2 f(E_+) + u_k^2 f(E_-)\right] = n_b$$

$$n_a + n_b = \frac{1}{N}\sum_k\left[f(E_+) + f(E_-)\right]$$

---

again:

def: (1) $\varepsilon_a(k)$, $\varepsilon_b(k)$ (recimo, dano)

(2) (init n_a, n_b) $\bar\varepsilon_a = \varepsilon_a(k) + V n_b$, $\bar\varepsilon_b = \varepsilon_b(k) + V n_a$ $\Rightarrow$ $\eta_k = \frac{\bar\varepsilon_a + \bar\varepsilon_b}{2} - \mu$?, $\xi_k = \frac{\bar\varepsilon_a - \bar\varepsilon_b}{2}$

(3) $E_\pm = \eta_k \pm \sqrt{\xi_k^2 + \Delta^2}$ (init Δ)

(4) za init. param: const. $= n_a + n_b = \frac{1}{N}\sum_k \left(f(E_+) + f(E_-)\right) \to$ določi μ.

1' half f.

(5) gap eq:

$$\Delta' = \frac{V}{2}\Delta\cdot\sum_k \frac{f(E_-) - f(E_+)}{\sqrt{\xi_k^2 + \Delta^2}} \to \text{update } \Delta \to \Delta' \quad (E_\pm \to E_\pm')$$

(6) update n_a, n_b: • izračunaj $u_k'^2 = \left(1 + \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta'^2}}\right)$, $v_k'^2 = \left(-\,'' - \text{z minusom}\right)$

• $n_a' = \frac{1}{N}\sum_k \left[u_k'^2 f(E_+') + v_k'^2 f(E_-')\right]$, podobno n_b'

$n_a \to n_a'$, $n_b \to n_b'$ (tudi $\eta_k \to \eta_k'$, $\xi_k \to \xi_k'$)

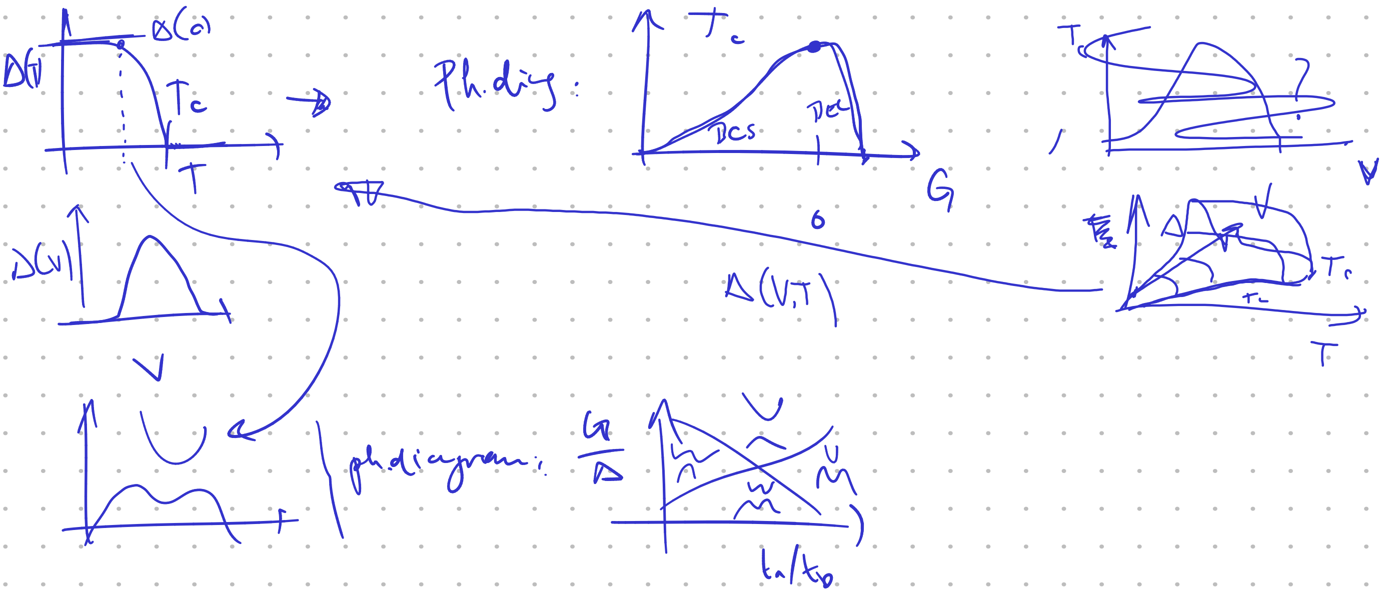
(7) • update μ' $\Leftarrow$ $n_a' + n_b' = \text{const.}$

$\mu \to \mu'$

(8) nazaj v gap eq. $\Delta'' = \ldots$

2 dim tight binding toy model: N sites, PBC

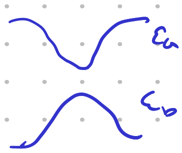
Vprašanji: • U_H = nonc vs $U_{\sigma}=0$? kaj je to si Hartree shifter?
• Fig 5: zakaj je μ/t_a vs V/t_a vzorčni?



state: $\vec{X} = (\Delta, m, \mu)$, $m = n_a - n_b$, $n = n_a + n_b = 1$ (oz. const./fixed)

2D tight binding, dve orbitale: $\varepsilon_a(\vec{k}) = \frac{D}{2} - 2t_a(\cos k_x + \cos k_y)$, $t_a = 1$

$\varepsilon_b(\vec{k}) = -\frac{D}{2} - 2t_b(\cos k_x + \cos k_y)$



$w_k = \frac{1}{N_k} = \frac{1}{N^2} \to \frac{1}{N_k}\sum_k f(k)$ $n_a = \frac{n+m}{2}$, $n_b = \frac{n-m}{2}$

↳ $m=0$ → enako zasedeni
$m>0$: a več elektronov
$m<0$: a manj el.
$|m|\sim 1$: močna polarizacija

→ $\bar{\varepsilon}_a = \varepsilon_a + V n_b$
$\bar{\varepsilon}_b = \varepsilon_b + V n_a$ → $h_k = \begin{pmatrix} \bar{\varepsilon}_a - \mu & -\Delta \\ -\Delta & \bar{\varepsilon}_b - \mu \end{pmatrix}$ → $\eta_k = \frac{\bar{\varepsilon}_a + \bar{\varepsilon}_b}{2}$, $\xi_k = \frac{\bar{\varepsilon}_a - \bar{\varepsilon}_b}{2}$, $E_k^{\pm} = \eta_k - \mu \pm E_k$
(delta)

→ $\Delta = V\Delta \sum_k w_k \frac{f(E_-) - f(E_+)}{2E_k}$

→ $\vec{R} = \begin{pmatrix} \Delta - \Delta' \\ m - m' \\ \mu - \mu' \end{pmatrix}$ → $\underline{\vec{R} = 0 \text{ rešitev}}$

pair

$$\cancel{1} = \frac{V}{2}\sum_k \frac{1}{N_k} \frac{f(E_-) - f(E_+)}{|\xi_k|}$$ e okoli $\Delta = 0$

$\lambda(V,T)$

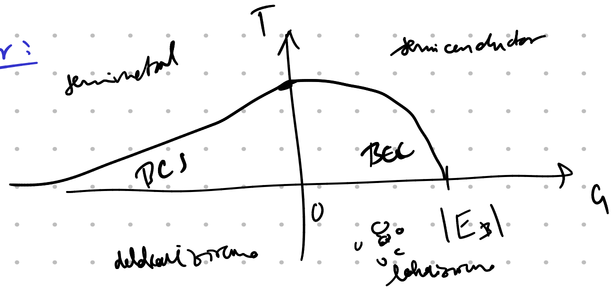
→ $\lambda < 1$ → stabilno $\lambda = 1$ → ravna meja

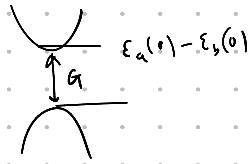
$\lambda > 0$ → nestabilno za $\Delta > 0$.

o) parameters → reši za (m, μ) pri $\Delta = 0$. → search $\Delta > 0$, če obstaja, če $\exists\, \Delta > 0$, potem primerjaj energijo pri $\Delta = 0$ in $\Delta > 0$, vzami stabilno.

opomba: $\overline{\varepsilon_a} - \overline{\varepsilon_b} = \varepsilon_a - \varepsilon_b - Vm$ → $\xi_k \propto -Vm$ → če V velik → $\Delta \to 0$.

BCS – BEC crossover:





če imamo [illegible] pasove → excitons z com momentom $\vec q$ ⇒ $\Delta_{\vec q} \propto \langle a^\dagger_{\vec k+\vec q}\, b_{\vec k}\rangle$.

Ana magistrska zapiski:

stabilnost excitona:

$$\varepsilon_{a,b} = \varepsilon_{a,b} \mp \frac{\hbar^2 k^2}{2m_{a,b}} \qquad \frac{1}{m_e} = \left(\frac{1}{m_a} + \frac{1}{m_b}\right)$$

→

$$\left(-\frac{\hbar^2}{2m_e}\frac{\partial^2}{\partial \vec r^2} + \frac{e_0^2}{4\pi\varepsilon_0 \varepsilon|\vec r|}\right)\psi(\vec r) = \underbrace{E}_{-E_b}\psi(\vec r) \qquad \psi(r) \simeq \sum_k \chi_k e^{ikr}$$

$$\to \left[\left(\frac{\hbar^2 k^2}{2m_e} + E_b\right)\right]\chi_k = -\sum_{k'} V_{k-k'}\chi_{k'} \qquad \chi_k = -\frac{\Delta_k}{2E_k}$$

$$\Delta_k = \sum_{k'} \frac{V_{kk'}\Delta_{k'}}{2E_{k'}} = -\sum_{k'} V_{kk'}\chi_{k'} = -\chi_k 2E_k = -\sqrt{\xi_k^2 + 4\Delta_k^2}\,\chi_k$$

$$\sqrt{(\varepsilon_a-\varepsilon_b)^2 + 4\Delta_k^2} = \frac{\hbar^2 k^2}{2m_e} + E_b \to (|E_g| - E_b)\underbrace{\left(\frac{\hbar^2 k^2}{2m_e} + E_g + E_b\right)}_{>0} + \underbrace{4\Delta_k^2}_{\text{če } =0} = 0$$

$$\left(E_g + \frac{\hbar^2 k^2}{2m_e}\right)^2$$

$$\Downarrow$$

$$E_g = E_b$$

brezdimenzijsko: $\varepsilon_{ex} = \frac{\hbar^2}{m r_{ex}^2}$

$$\frac{\varepsilon_a(k)}{\varepsilon_{ex}} = \frac{E_g}{2\varepsilon_{ex}} + \frac{(k r_{ex})^2}{2\left(\frac{m_a}{m_0}\right)} = \frac{E_g}{2} + \frac{k^2}{2m_a}$$

→ $\boxed{m_0,\ \varepsilon_{ex},\ r_{ex} \text{ so skale}}$

→ $E_n = -\frac{m_{ex}}{m_0\varepsilon^2} Ry \cdot \frac{1}{n^2}$ → max: $E_1 = -E_b$, $r_{ex} = \frac{m_0\varepsilon}{m_{ex}} a_B = \frac{Ry}{\varepsilon E_b} a_B$.

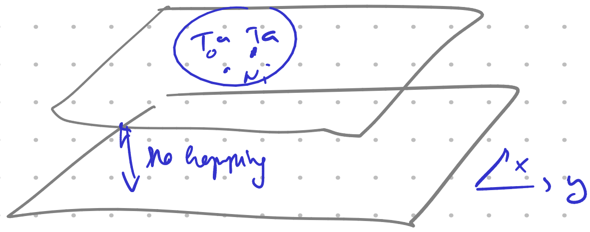
• model Ta_2NiSe_5:

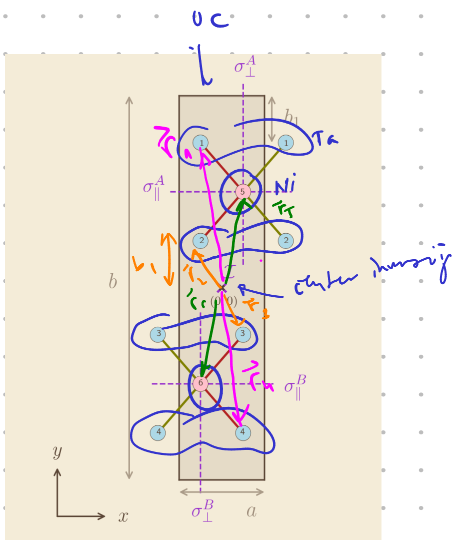
→ 6 Wannier orbital v O.C.

↳ 4 Ta (5d)

↳ 2 Ni (3d)

↳ Se ne upoštevamo neposredno.





$$\vec r_1 = -\frac{a}{4}\vec e_x + \left(\frac{b}{2} - b_1\right)\vec e_y = -\vec r_4$$

$$\vec r_2 = -\frac{a}{4}\vec e_x + b_1 \vec e_y = -\vec r_3$$

$$\vec r_5 = \frac{a}{4}\vec e_x + \frac{b}{4}\vec e_y = -\vec r_6$$